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Synergistic maturity group and planting date adaptation sustains soybean yields under warming

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Warmer summers can decrease U.S. soybean seed yields if agronomic management is not adapted accordingly. Our research question is: Can farmers sustain soybean yields under high temperatures by changing cultivar maturity group and sowing dates? We quantified the impacts of high temperatures on soybean yields across thousands of fields in the US Mid-South and investigated whether adapting sowing dates and maturity groups could offset these negative impacts. Soybean yields dropped by nearly 6% on average with a +2Â°C temperature increase. However, planting earlier and choosing a later-maturing soybean variety can increase seed yields by 12% on average under high temperatures. Maturity group selection and planting date are among the most actionable management decisions for farmers, as they can rapidly adopt these changes without incurring significant production costs. These choices can be further optimized under warming to offset yield penalties.

SYNERGISTIC MATURITY GROUP AND PLANTING DATE ADAPTATION SUSTAINS SOYBEAN YIELDS UNDER WARMING

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ABSTRACT

High temperatures increasingly threaten U.S. soybean seed yields, yet actionable solutions remain limited. We evaluated the APSIM model against observed regional data from the U.S. Mid-South and conducted gridded simulations across 4,651 fields to identify how concurrent changes in planting date and maturity group affect yield under rising temperatures and elevated [CO_2]. The +2°C temperature scenario reduced yields by 5.7% on average. However, combining a later-maturing soybean genotype with earlier sowing synergistically increased soybean yield by 12% under the +2°C scenario. These results revealed an opportunity to sustain soybean yield through management and genotype selection under warming. Advancing our understanding of management and genotypic effects on soybean phenology and improving their representation in process-based models are therefore essential for developing adaptive strategies that can support durable yield gains under high-temperature conditions.

1 INTRODUCTION

Rising temperatures and changes in rainfall patterns can decrease U.S. soybean seed yields by 19% in the absence of adaptive strategies (Elli et al., 2022). Higher temperatures accelerate phenological development, shortening the crop cycle and limiting biomass accumulation and yield (Elli et al., 2022; Rose et al., 2016). Previous simulation studies indicate that adjusting soybean phenology may help offset yield reductions under global warming (Yang et al., 2025; Zabel et al., 2021). While projected increases in [CO2] can partially offset heat-induced yield losses (Ainsworth et al., 2002a; Kothari et al., 2022), complementary agronomic adaptation may be needed in some environments to fully offset the yield penalty (Makowski et al., 2020).

A relatively simple and well-established strategy to adjust the soybean life cycle to the available growing season length is to select soybean maturity groups based on local temperature and photoperiod conditions (Mourtzinis and Conley, 2017). Moreover, earlier planting extends the growing season and shifts the crop cycle into cooler periods, reducing exposure to late-season drought and heat events (Edwards et al., 2003; Bastidas et al., 2008). While farmers have been shifting towards earlier planting and maturity groups well-adapted to a region, the synergistic effect of these shifts under a warming scenario remains unquantified.

We conducted a comprehensive simulation study across 4,651 fields and 40 weather-years using a well-tested version of the APSIM (Agricultural Production Systems Simulator) framework. We considered soybean production systems in the Arkansas Delta region as a case study, an environment particularly vulnerable to warming, with hot summers and frequent late-season drought events, which accounts for over 1.5 million hectares of soybean-planted area (NASS, 2025). We hypothesize that under high temperatures, using late-maturing soybean genotypes combined with early planting will offset the negative impacts of warming on soybean yield. Our objective was to quantify the individual and synergistic impacts of changes in maturity group and sowing date on adapting soybean production to high-temperature risks under current and elevated [CO_2].

2 MATERIAL AND METHODS

2.1. APSIM-Soybean model testing

The soybean model within APSIM Next Generation (Holzworth et al., 2018) was evaluated against regional data from the U.S. Midsouth covering multiple locations, planting dates, and cultivar maturity groups (Figure S1), as described in Salmerón et al. (2016). The APSIM-Soybean model has previously proven successful in simulating the growth of various soybean maturity groups under different environmental conditions in the US (Archontoulis et al., 2020; Elli et al., 2022). We further tested the model against experimental data and updated cultivar coefficients to the latest APSIM Next Generation (Holzworth et al., 2018) version 2025.3.7681. Cultivar coefficients were calibrated for an early maturity group (MG) 4 (relative MG 4.0 to 4.4) and an early MG 5 (rMG from 5.0 to 5.4) cultivar. The calibration results confirmed that the model is robust across a wide range of environmental conditions (Figures S1 and S2 and Table S1), with root mean square errors within similar ranges to those reported in other studies for soybean phenology and yield (Archontoulis et al., 2014; Salmerón and Purcell, 2016; Salmerón et al., 2017).

2.2. Regional simulations and scenario analysis

We conducted a simulation exercise across 40 weather years and 4,651 fields in the US Mid-South. Further information on the weather conditions of the study area is presented in Figure S3. We used the soybean cropland data layer (NASS, 2025) to identify soybean-cropped pixels and define the spatial extent of the simulation fields, which were resampled to a 2.5 km grid (~0.025°). Each grid cell classified as soybean cropland was treated as one simulation field. We used the *apsimx* R package (Miguez, 2020) to run the simulations. Daily historical weather data on maximum and minimum temperatures, solar radiation, and rainfall were retrieved from the nearest weather station in the National Weather Service Network using the *apsimx* R package. [CO2] was set at 350 ppm. Soil profile information was derived from SSURGO (Soil Survey Staff, 2023). Within each simulated grid cell, we used the most predominant soil profile. Soil water and inorganic nitrogen initial conditions were reset on January 1 each year and were held constant across locations to avoid confounding effects of soil vs. weather (Elli and Archontoulis, 2023). On the first day of the simulation (January 1), soil water content was set at field capacity, total inorganic nitrogen over the soil profile was 50 kg N ha^{-1}, and maize residue on the soil surface was 2000 kg ha^{-1} with a C:N ratio of 70.

For baseline simulations, the sowing date was May 22, an average value for the studied region based on USDA-NASS 50% sowing progress from 1990 to 2025 (NASS, 2025; Figure S4). We used an early MG 4 cultivar, which is well adapted to the region (Salmerón et al., 2016; Salmerón et al., 2014). Soybean was irrigated as it is the predominant system across the studied region (NASS, 2025). Automatic irrigation was triggered between R1 and R7 when soil water reached 50% of the plant-available water in the 0-60 cm depth, to refill soil moisture to field capacity.

For the high-temperature scenario, we considered a 2°C increase (+2°C) in daily maximum and minimum temperatures, which is an average projection derived from 15 global circulation models for 2050, as reported by Elli et al. (2022). A +2°C scenario combined with an elevated [CO2] to 540 ppm (Elli and Archontoulis, 2023) was also evaluated. We employed a simpler approach, rather than the direct outputs of global circulation models, because it has been shown to yield results with similar direction and magnitude while substantially reducing computational requirements (Elli et al., 2022). The adaptive scenarios included (1) an early MG 5 cultivar instead of early MG 4, (2) an early sowing date by 4 weeks (April 24), corresponding to the average USDA-NASS 10% sowing progress from 1990 to 2025 (NASS, 2025; Figure S4), and (3) a combination of an early MG 5 cultivar with early sowing. Data analysis and visualization were conducted using R version 4.1.1 (R Development Core Team, 2024).

3 RESULTS

3.1. Elevated temperature and [CO2] impacts on soybean yields without adaptation

The +2°C temperature scenario decreased soybean seed yields by 5.7% on average (~245 kg ha^{-1}), with values ranging from -14.7% to -2.5% across the 4,651 sites (Figure 1). The mean simulated yield for the baseline condition was 4,300 kg ha^{-1} with a south-to-north geographical gradient ranging from 2,142 to 4,837 kg ha^{-1} (Figure S5). Elevated [CO_2] consistently offset the negative effects of higher temperatures across sites (Figure 1), with an average yield 12.9% greater than under baseline conditions, and values ranging from 9.2% to 15.4% across all sites (Figure 1). A shortening of the total crop cycle was observed in 61% of the sites, with more pronounces changes in northern environments (Figure 2), where baseline temperatures are lower. The total crop cycle length decreased by up to 15 days across all site-years. The seed filling period was shortened by up to 9 days across all site-years.

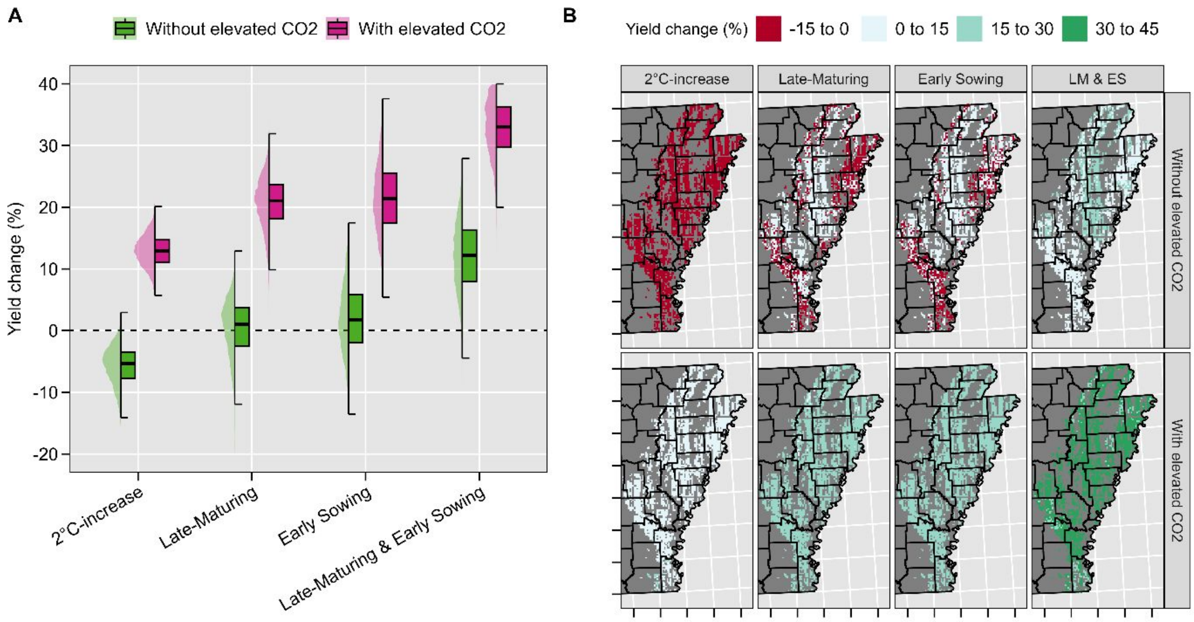

Figure 1. Overall relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on soybean seed yield across 4,651 soybean fields and 40 weather years in the US Mid-South (A), and their geographical distribution (B). Values per pixel in panel B are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C.

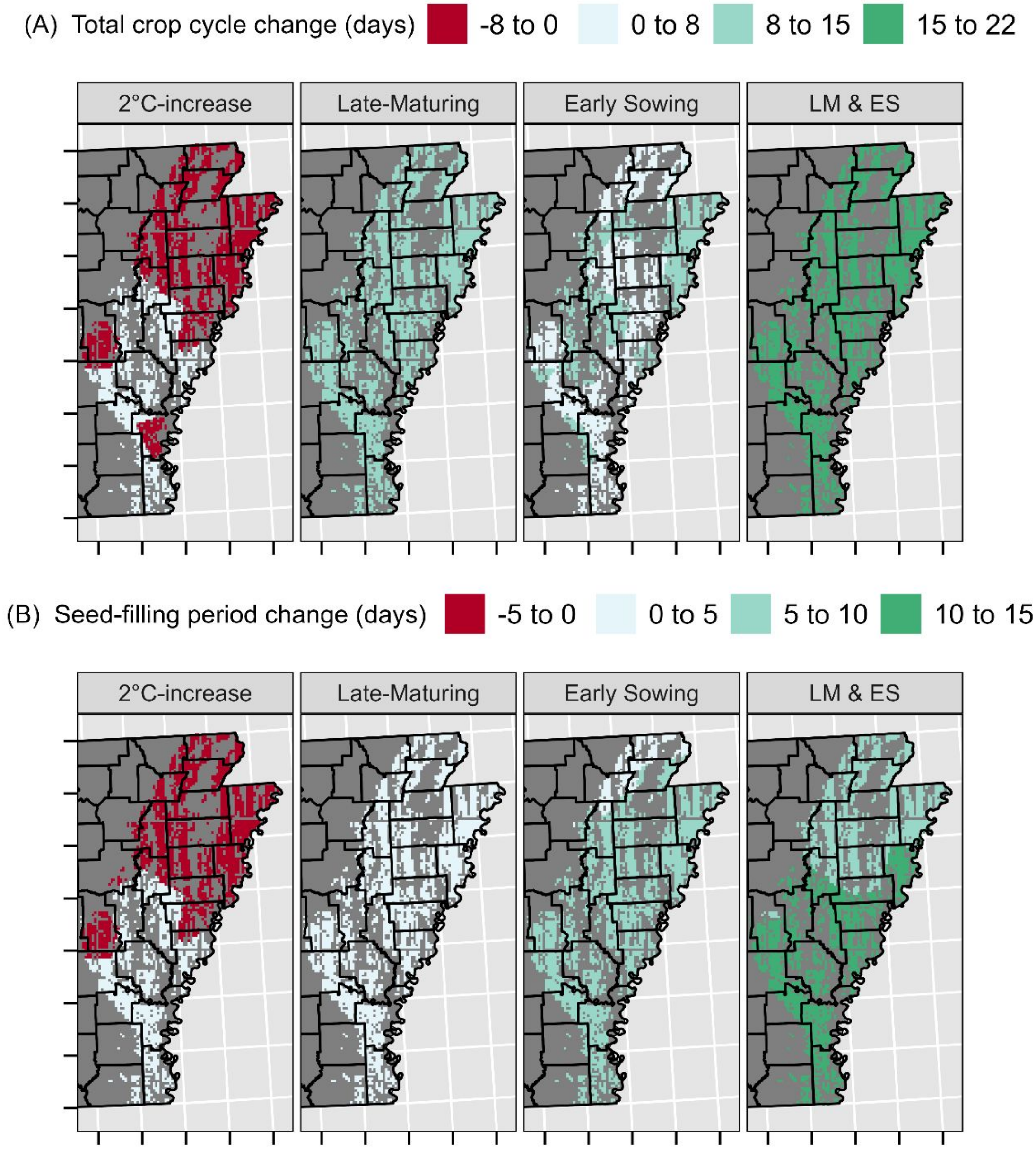

Figure 2. Geographical distribution of the relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on changes in soybean total crop cycle (A) and seed-filling period (B) across 4,651 soybean fields in the US Mid-South. Values per pixel are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C.

3.2. Adaptive strategies to elevated temperature

Stacked agronomic management strategies were more effective than standalone strategies to maximize soybean yields under warming (Figure 1). Using a late-maturing soybean cultivar in combination with early sowing increased soybean seed yield by 12% under the +2°C temperature scenario, without elevated CO_2. The combined adaptive scenario extended the seed-filling period by an average of 10.4 days (Figure 2). Overall, the standalone strategy of using a late-maturing cultivar under a +2°C temperature scenario resulted in yields comparable to the historical baseline, with values ranging from -12.6% to 4.8% across fields. This practice increased yields at 62% of the 4,651 simulated sites, particularly in the central-north region (Figure 1B). The Early sowing strategy increased yields by an average of 2% relative to the baseline, with values ranging from -4.9% to 7.8%. This practice increased yields at 71% of the sites. Elevated [CO_2] altered the magnitude but not the direction of the yield response to stacked adaptive management options. The combined impact of elevated CO_2 and stacked agronomic management strategies increased yields by an average of 34%, with values ranging from 26.1% to 41.8% (Figure 1).

4 DISCUSSION

Our study revealed that soybean yields under warming can be sustained by concurrently adjusting cultivar maturity and planting date. Under the +2°C temperature scenario, the standalone strategies of using a late-maturing cultivar and planting earlier increased yield by an average of 0.5% and 2%, respectively, relative to the historical simulations. This implies that individual strategies offset the impacts of higher temperature, resulting in yields comparable to baseline yields under current weather conditions. However, the offset was not consistent across all sites (Figure 1). The combined strategies increased seed yield by an average of 12% in relation to the baseline, with values ranging from 0.4% to 17.6% across sites, demonstrating a synergistic impact and emerging as the most promising approach to sustain soybean yield under a +2°C scenario. Our study extends previous work (Elli et al., 2022; Nendel et al., 2023) by quantifying the synergistic effect of agronomic management strategies to mitigate heat risks in irrigated soybeans at a fine spatial resolution in the U.S. Mid-South.

We found a 5.7% decrease in soybean yield due to rising temperatures, with values ranging from -14.7% to -2.5% across sites (Figure 1). Elli et al. (2022) reported an average 19% reduction in seed yield at 10 locations in the US Corn Belt under a +2°C temperature scenario without adaptive strategies. While Elli et al. (2022) did not quantify the combined impact of high temperatures and agronomic management, they found that sowing 12 days earlier increased soybean seed yield by 4.7%. Deryng et al. (2014) investigated the effects of heat stress on soybeans worldwide and predicted a 33% yield reduction by 2080. Nendel et al. (2023) projected that heat stress will increasingly outweigh drought stress as a constraint to soybean production in Europe.

While elevated [CO_2] can offset the negative impacts of rising temperatures (Ainsworth et al., 2002b), genotypic differences in [CO_2] response may exist (Bishop et al., 2015). Moreover, model uncertainties exist in predicting yield benefits under elevated [CO_2] (Kothari et al., 2022). Recognizing that soybeans are already a highly heat-tolerant crop, heat events above 28°C during the reproductive phase can reduce yield by lowering seed number and weight (Boote et al., 2005; Djanaguiraman et al., 2013). The APSIM-soybean version used in the present study accounts for temperature effects on crop growth rate (Kothari et al., 2024), but does not account for the impacts of extreme heat on seed number determination, a common limitation in process-based models (Richetti et al., 2025). Moreover, APSIM assumes constant temperature responses throughout the life cycle, whereas in soybeans this response may change across developmental stages (Hatfield et al., 2011).

Warmer springs and longer frost-free periods have allowed farmers to plant earlier every year (Mourtzinis et al., 2019). However, this practice is not always feasible due to extreme rainfall events, which are projected to intensify further (Elli et al., 2022). Given the importance of adapting the soybean life cycle to temperature changes, further research is needed to enhance the fundamental understanding of soybean phenology and the genetic and management controls. Extending the seed filling period was a promising strategy to maximize soybean seed yield (Figure 1), which agrees with the findings of Boote et al. (2003). However, phenology did not fully explain yield reduction under warming, especially in the Southern region (Figures 1 and 2). This could be associated with slower development on days when temperatures exceed the optimum temperature for soybean development in APSIM (30°C, Figure S3). Battisti et al. (2018) observed that, owing to the temperature response function governing phenological development, the APSIM model simulated a greater reduction in soybean cycle length under a +4.5°C warming scenario than under a +6.0°C scenario in Southern Brazil, with a mean air temperature 23.65°C. Further yield decreases were likely associated with high-temperature effects on radiation use efficiency (Kothari et al., 2022; Ruiz-Vera et al., 2013). While exploring optimum planting date windows for target environments was not the focus of this study, future research could investigate more extreme scenarios to enhance mechanistic understanding of yield responses under warming.

5 CONCLUSION

This study advances our understanding of synergistic agronomic management adaptation for sustaining soybean seed yields under warming, providing new quantitative evidence from the irrigated Soybeans in the US Mid-South that extends the existing literature. We found that soybean yields declined by 5.7% on average under rising temperatures. Adjusting the soybean maturity group, combined with early sowing, consistently prevented yield losses during high temperatures, resulting in an overall yield gain of 12% relative to the historical baseline.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

AUTHOR CONTRIBUTIONS

Elvis F. Elli: Conceptualization, Data curation, Formal analysis, Funding acquisition, Visualization, Writing - original draft; Caio dos Santos: Formal analysis, Methodology Software Writing - review & editing; Samuel B. Fernandes: Methodology, Software, Writing - review & editing; Montserrat Salmerón: Investigation, Methodology, Resources, Writing - review & editing; Rafael Battisti: Methodology, Writing - review & editing; Rogério de S. Nóia-Júnior: Conceptualization, Investigation, Writing - review & editing.

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SUPPLEMENTAL MATERIAL

Supplemental materials are included in this paper.

Figure S1. Locations of multi-environment trials used to collect phenology and yield data for soybean maturity groups 4 to 5 (A). Relationship between APSIM-simulated and observed day of year (DOY) for key phenological stages: emergence, beginning of flowering, beginning of pod development, beginning of seed filling, and physiological maturity (B). Relationship between APSIM-simulated and observed soybean yield at 13% grain moisture (C).

Figure S2. APSIM-simulated (lines) and measured (points) leaf biomass (green colour), seed biomass (black colour), and total biomass (yellow colour) in Fayetteville, Northwest Arkansas, for 2 different maturity groups (early 4 and early 5). Measured data were obtained from an experiment conducted in 2008 under irrigated conditions in a randomized complete block design with 4 replications. Further information on field data collection can be found in (Mastrodomenico and Purcell, 2012). Vertical lines represent the standard deviations. RMSEs for seed yield, leaf biomass, and total biomass were 659 kg/ha, 572 kg/ha, and 1335 kg/ha, respectively.

Figure S3. Environmental characterization of baseline conditions across all studied site-years. The APSIM phenology response curve (0-1) is presented, where 1 represents maximum development rate. The solid vertical and horizontal lines indicate the median growing-season average temperature and total growing-season precipitation, respectively.

Figure S4. USDA-NASS soybean sowing progress in Arkansas progress from 1990 to 2025. Grey lines represent the historical years, while the green, thicker line indicates the average sowing progress. 50% and 10% average sowing progress are indicated in the figure.

Figure S5. Density distribution of simulated soybean yields without adaptation under the baseline and +2°C scenarios with and without elevated [CO2] (A), and its geographical distribution (B).

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Supplementary Material

SYNERGISTIC MATURITY GROUP AND PLANTING DATE ADAPTATION SUSTAINS SOYBEAN YIELDS UNDER WARMING

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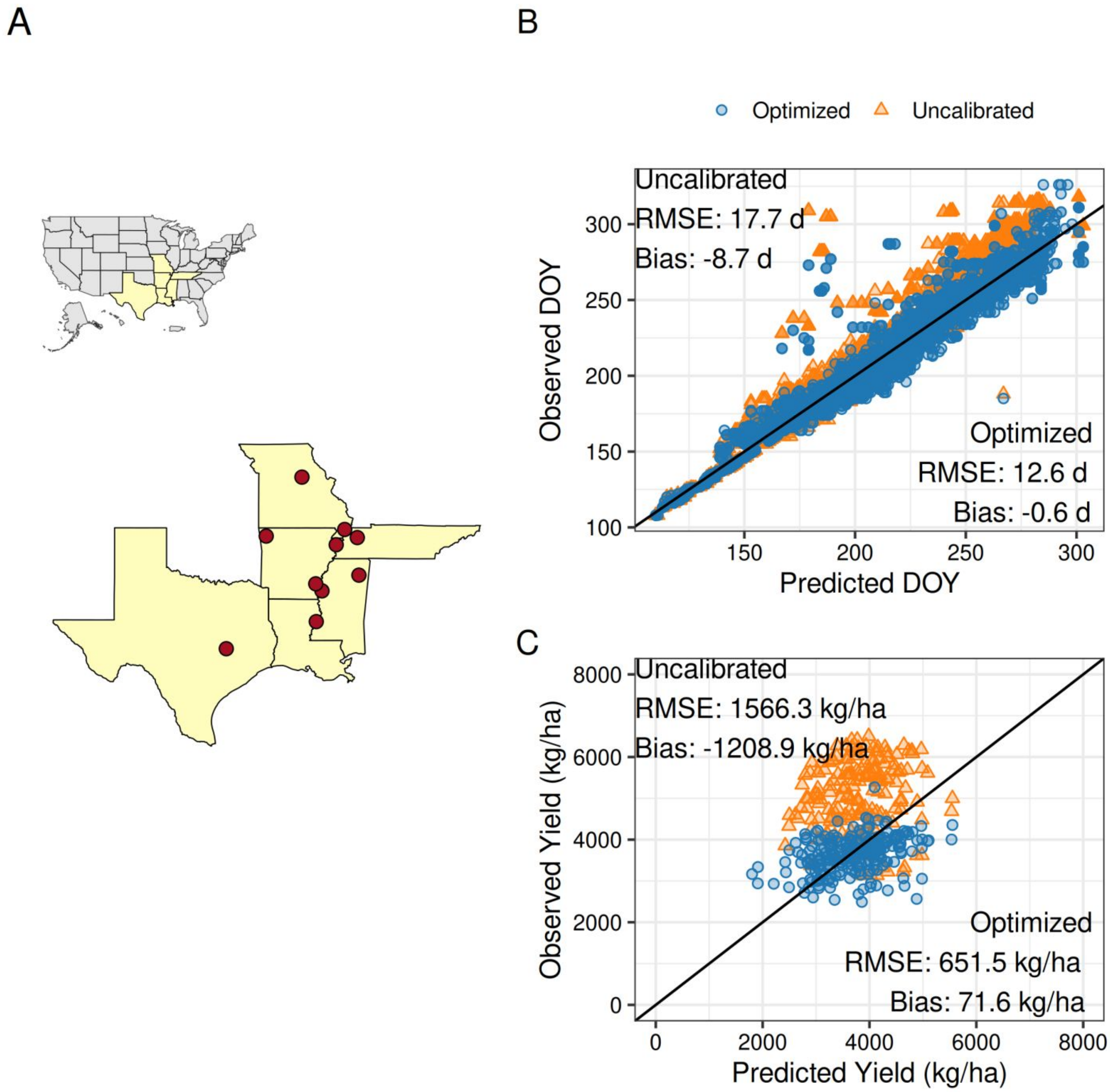

Figure S1. Locations of multi-environment trials used to collect phenology and yield data for soybean maturity groups 4 to 5 (A). Relationship between APSIM-simulated and observed day of year (DOY) for key phenological stages: emergence, beginning of flowering, beginning of pod development, beginning of seed filling, and physiological maturity (B). Relationship between APSIM-simulated and observed soybean yield at 13% grain moisture (C).

Model calibration

The occurrence of phenological stages, converted to days of the year (DOY), was used to derive the cultivar-specific model parameters (Table S1) using an iterative optimization procedure based on the Nelder–Mead algorithm (Nelder and Mead, 1965). This algorithm explores the parameter space iteratively using a geometric figure called simplex. The optimization procedure aims to minimize the residual sums of squares (Wallach et al., 2001) between the estimated and actual timing to reach the studied phenology stages for each maturity group (early 4 and early 5). We ran the algorithm using R software (R Development Core Team, 2021) and the APSIMX package (Miguez, 2020). Our focus was on optimizing phenology parameters while keeping cardinal temperature-related parameters constant, giving little evidence of changes across genotypes (Archontoulis et al., 2014). Two growth parameters were selected: area of the largest leaf and radiation use efficiency. The optimization workflow followed an approach similar to that outlined in Wallach et al. (2019), in which all parameters were optimized simultaneously to potentially capture parameter interactions.

Table S1. Maturity group (MG) – specific default and optimized APSIM parameter values targeted in the optimization of early and late maturity groups 4 and 5 cultivars.

APSIM Parameter	Units	Early MG 4	Early MG 5
Vegetative Target	°C-day	484	634
Early Flowering Target	°C-day	185	189
Early Grain Filling Target	Fraction (0-1)	0.338	0.042
Late Grain Filling Target	°C-day	384	418
Reproductive Photoperiod Modifier	Fraction (0-1)	11.48, 17.24	12.70, 14.31
Area of Largest Leaf	m^2	0.004	0.004
Radiation Use Efficiency	g MJ^{-1}	1.48	1.05

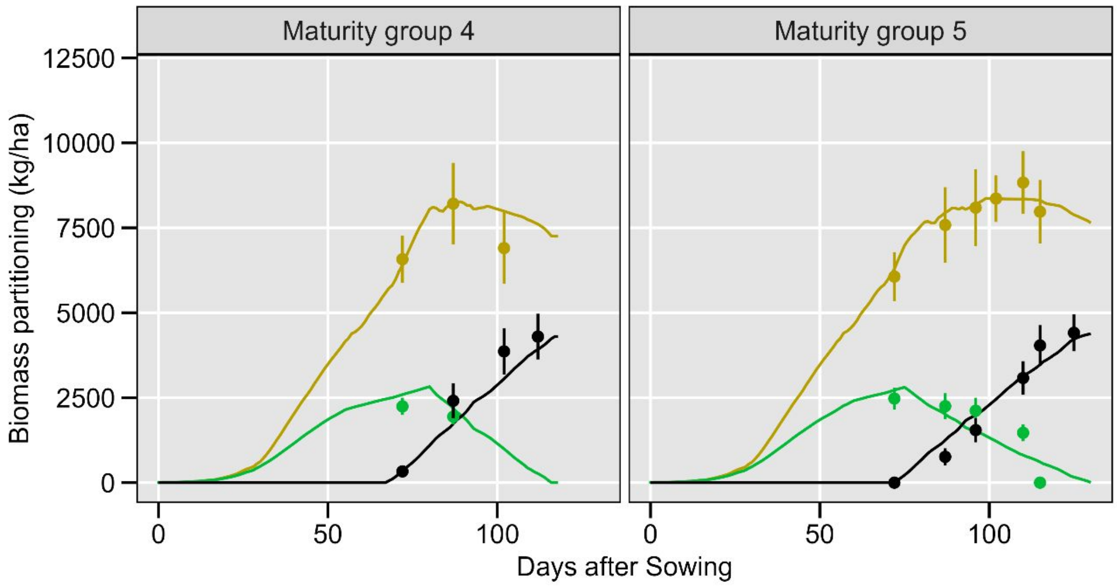

Figure S2. APSIM-simulated (lines) and measured (points) leaf biomass (green colour), seed biomass (black colour), and total biomass (yellow colour) in Fayetteville, Northwest Arkansas, for 2 different maturity groups (early 4 and early 5). Measured data were obtained from an experiment conducted in 2008 under irrigated conditions in a randomized complete block design with 4 replications. Further information on field data collection can be found in (Mastrodomenico and Purcell, 2012). Vertical lines represent the standard deviations. RMSEs for seed yield, leaf biomass, and total biomass were 659 kg/ha, 572 kg/ha, and 1335 kg/ha, respectively.

Study location and weather

The studied region is predominantly irrigated, with accumulated rainfall of 590 mm, unevenly distributed over the growing season (May to October). The average growing season temperature is 24°C, with values ranging from 20.9 to 27.1°C across all site-years (Figure S1). The studied area exhibits significant variability in soil conditions, ranging from shallow (less than 50 cm in depth) sandy soils with low soil water-holding capacity (0.4 mm cm^{-1}) to deeper soils with an effective root depth greater than 100 cm and high soil water-holding capacity (2 mm cm^{-1}). The soil organic carbon (0-30 cm) typically ranges from 0.5 to 3%.

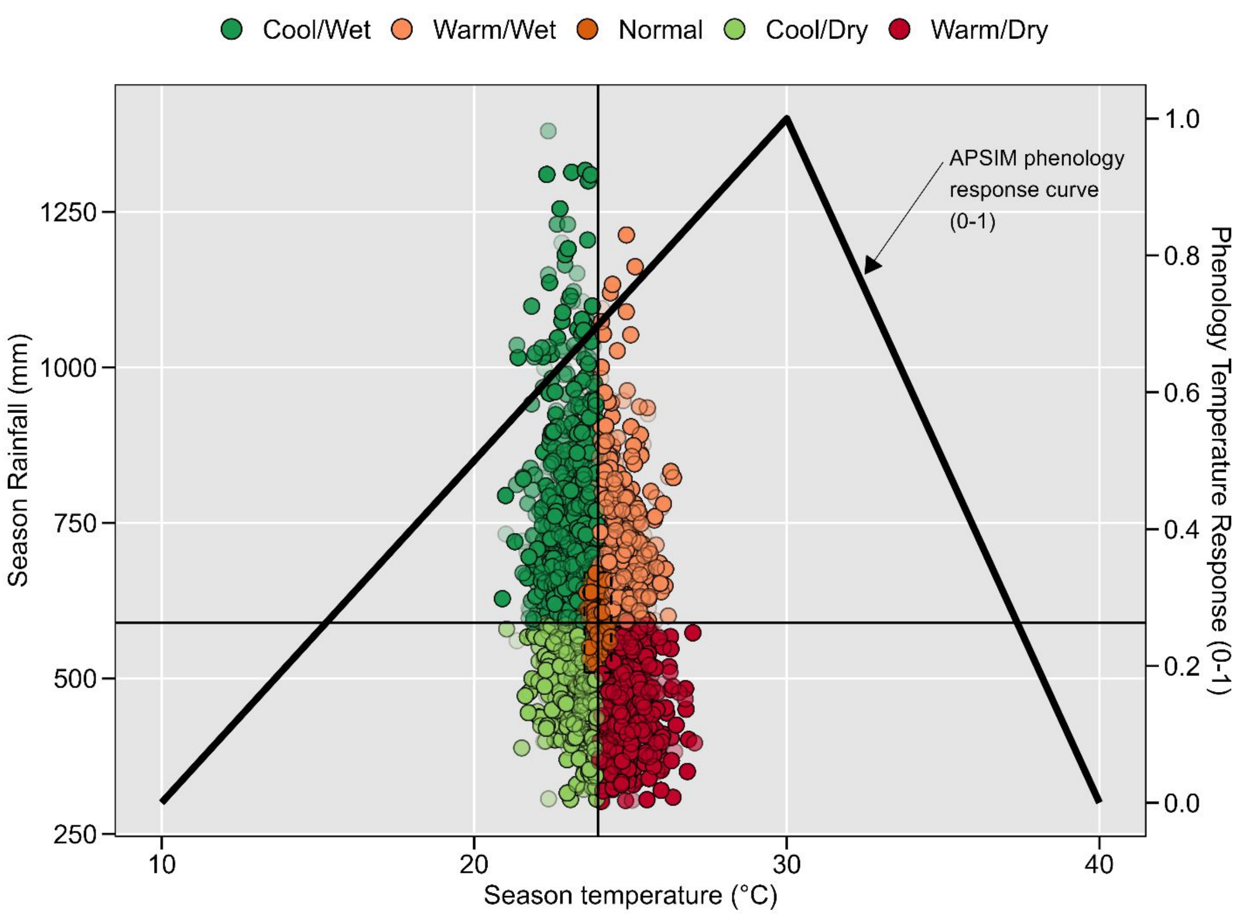

Figure S3. Environmental characterization of baseline conditions across all studied site-years. The APSIM phenology response curve (0-1) is presented, where 1 represents maximum development rate. The solid vertical and horizontal lines indicate the median growing-season average temperature and total growing-season precipitation, respectively.

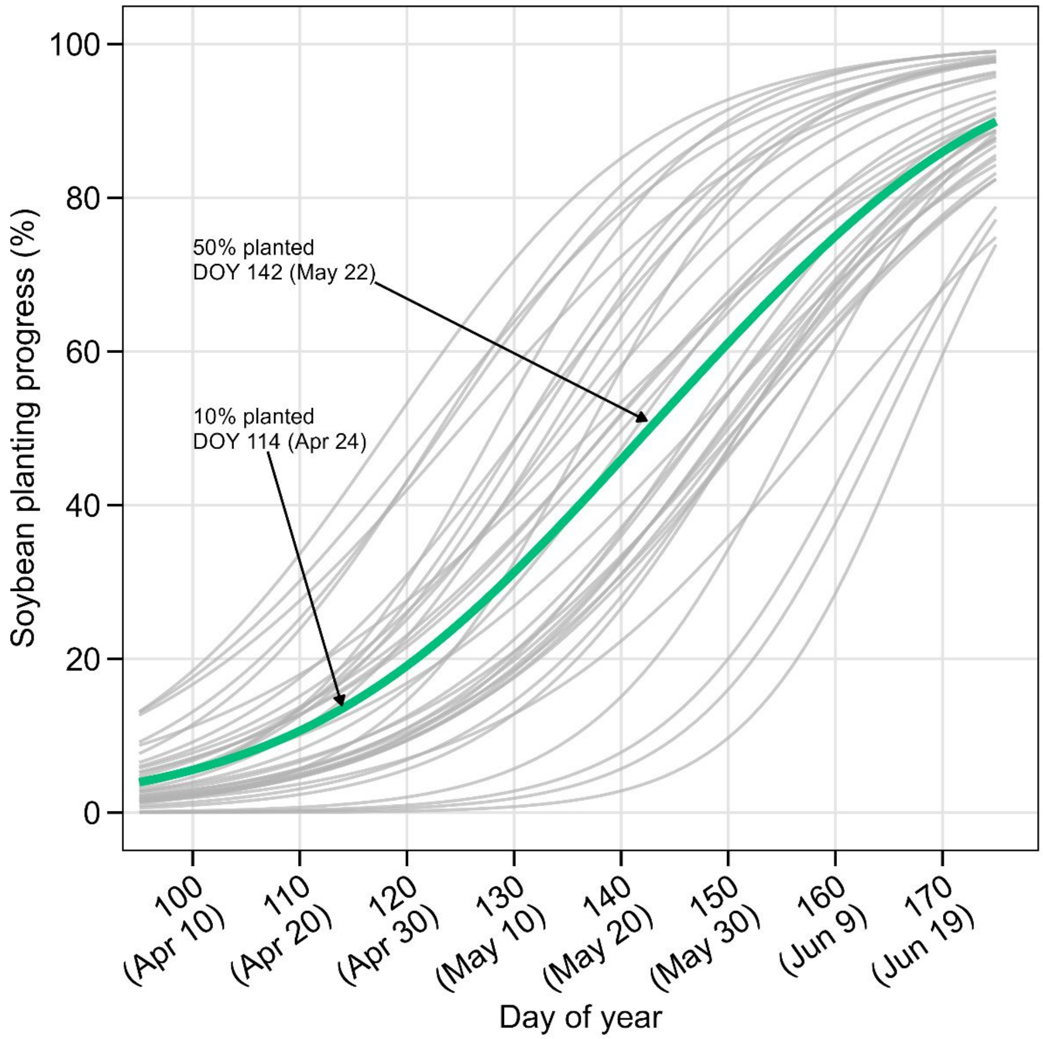

Figure S4. USDA-NASS soybean sowing progress in Arkansas progress from 1990 to 2025. Grey lines represent the historical years, while the green, thicker line indicates the average sowing progress. 50% and 10% average sowing progress are indicated in the figure.

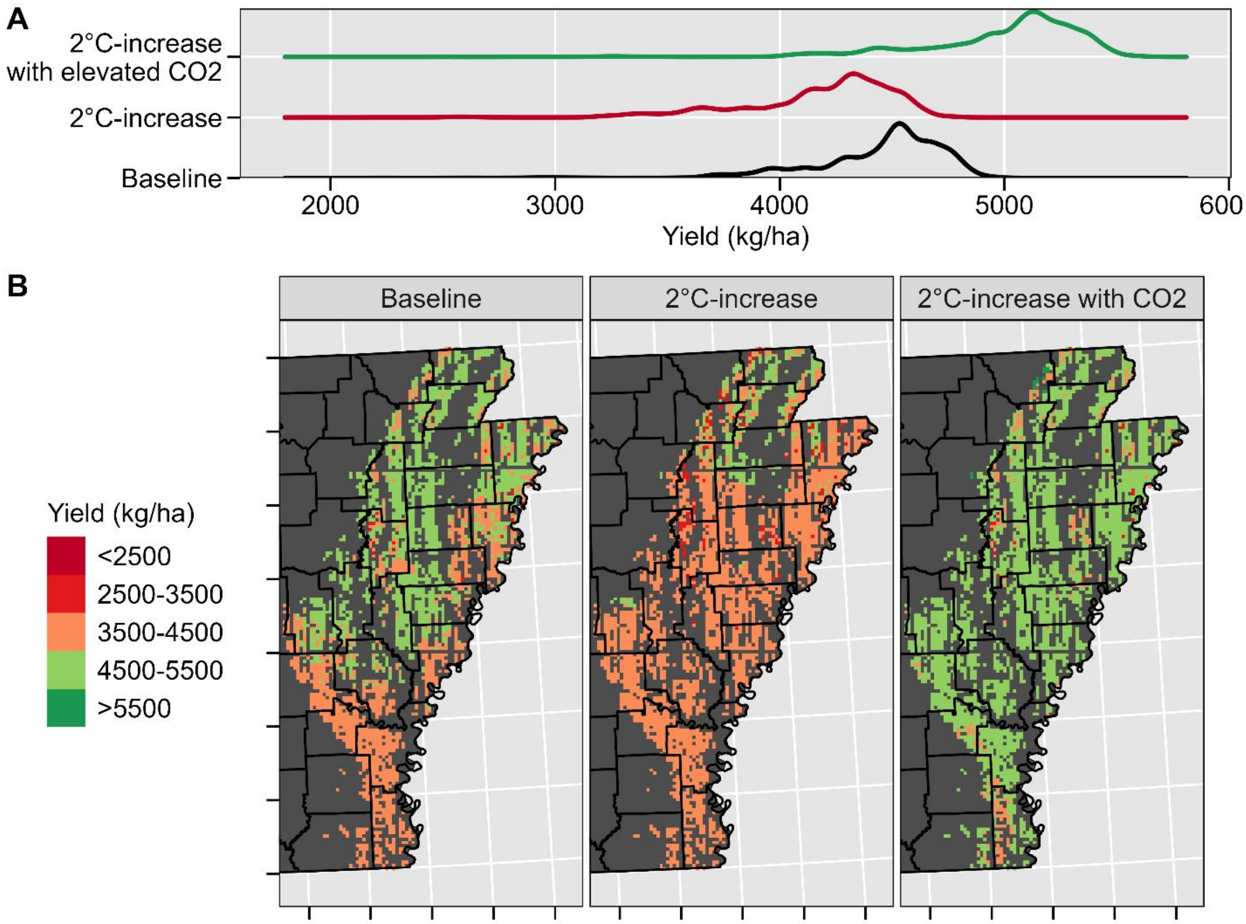

Figure S5. Density distribution of simulated soybean yields without adaptation under the baseline and +2°C scenarios with and without elevated [CO2] (A), and its geographical distribution (B).

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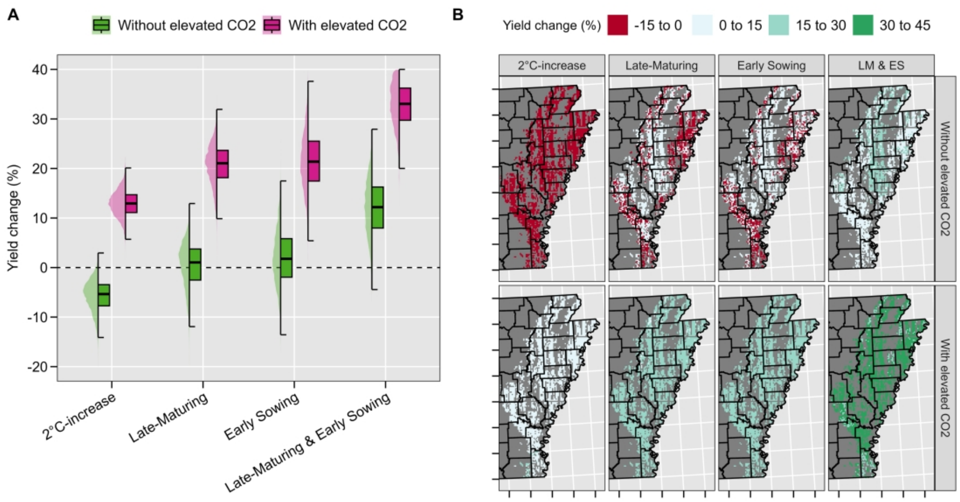

Figure 1. Overall relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on soybean seed yield across 4,651 soybean fields and 40 weather years in the US Mid-South (A), and their geographical distribution (B). Values per pixel in panel B are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C.

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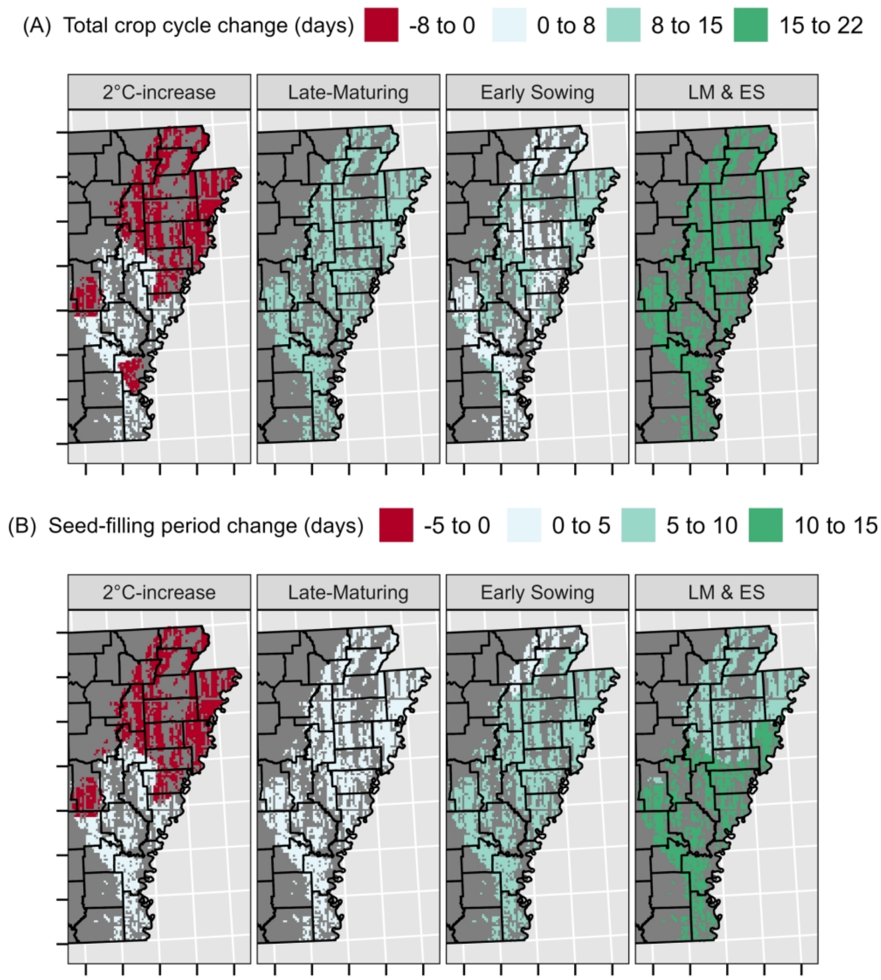

Figure 2. Geographical distribution of the relative effects of rising temperatures and adaptive genotypic and agronomic management strategies on changes in soybean total crop cycle (A) and seed-filling period (B) across 4,651 soybean fields in the US Mid-South. Values per pixel are averages across 40 weather years. LM & ES represent the combined effect of a late-maturing variety and early sowing. All adaptive scenarios were simulated under +2°C.

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